

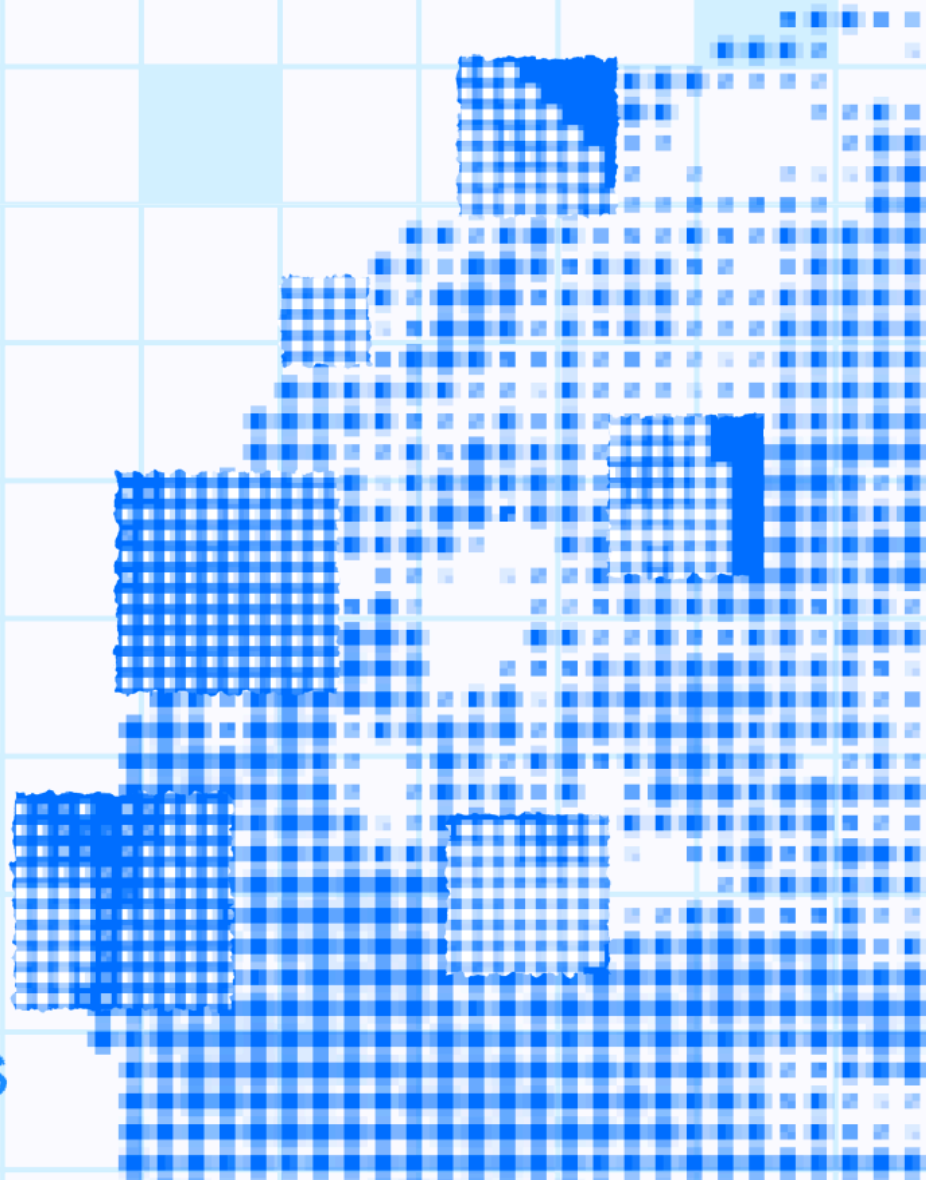
Optimized



AI-Powered Carbon-Aware Logistics Intelligence for
Resilient and Sustainable Delivery Networks

Routing
&
Carbon

Analytics



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Executive Summary Project

Indonesia's logistics sector loses *USD 6.1 billion annually* due to traffic congestion [11], while e-commerce demand reached *USD 65 billion GMV in 2024* and grow continuously [1][2]. However, many delivery decisions remain reactive and only made after disruptions occur. *ORCA (Optimized Routing and Carbon Analytics)* is an AI-powered logistics intelligence designed for Blibli to enable proactive decision-making. The platform combines a *LightGBM* delay prediction model (test F1-score: 0.31; AUC-ROC: 0.73), an *NSGA-II* multi-objective route optimization engine targeting *10–15% fuel savings* through *Pareto-optimal* routing, and an *SLA risk scoring* system that generates real-time alerts before delivery failures occur. A carbon footprint dashboard tracks *route-level emissions* using *GLEC v3.0*-aligned methodology [15], supporting Blibli's ESG reporting initiatives [9][10]. Drawing from McKinsey and WEF benchmarks [12][13][14], ORCA targets 20–30% fewer delivery delays, 10–15% better fleet utilization, 5–20% lower per-delivery costs, and 10–20% lower logistics carbon emissions.

Problem Statement

Selected Case Statement

LOGISTIC - BLIBLI

Selected Sub-Case Statement

AI-POWERED GREEN & RESILIENT LOGISTICS NETWORK

Main Objectives

ORCA aims to transform Blibli's logistics operations from reactive decision-making into a predictive, data-driven, and sustainability-aligned ecosystem. The platform leverages AI-powered delay prediction, *SLA risk scoring*, and *carbon-aware optimization* to improve delivery reliability while reducing operational and environmental costs simultaneously. McKinsey research on AI adoption in supply chains shows that early adopters improve logistics costs by up to 15% and service levels significantly compared to slower-moving competitors [12]. Building from this, ORCA targets a 20 to 30% reduction in delivery delays through *predictive analytics* using historical travel time, weather conditions, *hub dwell time*, traffic congestion, and driver behavior data. SLA compliance is projected to improve by at least 15% through proactive identification of at-risk deliveries before breaches occur. Operationally, AI-driven route optimization delivers 5 to 20% reductions in logistics costs [13] while fleet utilization improves 10 to 15% through intelligent dispatch and congestion-aware routing. On sustainability, McKinsey and WEF research shows AI-optimized routing reduces fuel usage by up to 15% [14], supporting a projected 10 to 20% reduction in logistics carbon emissions per routing decision. As an IDX-listed company, Blibli can also utilize ORCA to strengthen ESG transparency and sustainability reporting capabilities [9][10].

Problem Definition

1.1. What is the main problem?

Indonesia's logistics sector faces growing operational complexity due to rising distribution demand, particularly from e-commerce growth, while still being constrained by infrastructure limitations and environmental challenges. These issues often lead to delivery delays, reduced compliance with *Service Level Agreements* (SLAs), and inefficient fleet utilization across large multi-modal logistics networks.

At the national level, logistics costs in Indonesia remain among the highest in ASEAN, accounting for approximately 14.3% of Gross Domestic Product (GDP), or around USD 189 billion, indicating structural inefficiencies within the logistics system [3]. In addition to economic challenges, logistics operators are increasingly required to reduce fuel consumption and carbon emissions to meet sustainability targets. However, current logistics decision-making remains largely reactive and lacks predictive capabilities. The absence of AI-driven logistics intelligence systems capable of predicting SLA risks and optimizing routes based on speed, cost, and environmental impact contributes to persistent inefficiencies, increasing both operational costs and environmental burdens.

1.2. Who is impacted and to what scale?

The primary stakeholders impacted are Blibli's logistics operations, delivery partners, and millions of customers within the Blibli ecosystem. As nationwide order volumes continue to grow, logistics networks must coordinate deliveries across warehouses, fulfillment hubs, retail stores, and last-mile partners under increasingly demanding Same-Day and Next-Day SLA expectations. Operational inefficiencies in last-mile delivery directly increase fuel consumption, reduce fleet productivity, and weaken delivery reliability, ultimately affecting profitability and customer retention. Research shows that consumers stop purchasing from a retailer after experiencing a single late delivery [8]. At the corporate level, limited shipment-level carbon visibility also creates challenges for ESG reporting and sustainability target achievement. Without predictive logistics intelligence, businesses struggle to proactively manage operational risks, optimize resource allocation, and maintain customer trust at scale. These challenges impact not only logistics performance, but also long-term competitiveness and sustainability credibility [9][10].

1.3. Prove the problem

Indonesia's digital economy reached USD 90 billion GMV in 2024, with e-commerce contributing USD 65 billion and driving exponential demand for logistics capacity [1][2]. However, Indonesia ranked 61st out of 139 countries in the World Bank Logistics Performance Index 2023, dropping 17 places from its 2018 position, with the steepest declines recorded in timeliness and tracking reliability [5][6]. Average postal delivery times in Indonesia reached 13.3 days as recorded in LPI-referenced postal data, significantly exceeding Malaysia's 5.2 days and China's 5.6 days [6]. Indonesia's logistics costs remain among the highest in ASEAN at approximately 14.3% of GDP, signaling persistent structural inefficiencies in transportation and distribution that AI systems can directly address [3]. Simultaneously, the global transport sector produced nearly 8 Gt of CO₂ in 2022 [4], increasing ESG pressure on IDX-listed companies such as Blibli to demonstrate measurable sustainability outcomes [9]. McKinsey's 2024 consumer survey further shows that delivery reliability has overtaken speed as consumers' top priority, with on-time performance now ranking as the single most important factor in e-commerce delivery decisions [7].

Problem Solution

2.1. Main Solution

ORCA (*Optimized Route and Carbon Analytics*) is a logistics intelligence platform designed to enhance large-scale e-commerce operations such as Blibli by addressing key challenges in delivery performance, route efficiency, and sustainability monitoring. The platform proactively detects potential delivery issues by assigning delay probabilities and *SLA risk scoring* to active shipments, enabling operations teams to identify and address problems before they impact customers. It also applies *multi-objective optimization* using *NSGA-II* to evaluate route alternatives based on speed, cost, CO₂ emissions, and SLA risk, helping dispatchers make more informed decisions aligned with business priorities. In addition, ORCA provides shipment level carbon tracking by calculating emissions using the *GLEC v3.0* methodology, based on distance, load, and vehicle emission factors, which can then be aggregated into trends and breakdowns for ESG reporting. The system processes real-time shipment events through *Redis* queue, leverages *LightGBM* models for *predictive analytics*, and delivers operational insights via a *FastAPI* backend and *Next.js* dashboard with near real-time monitoring and alerting capabilities.

2.2. Attachment Diagram

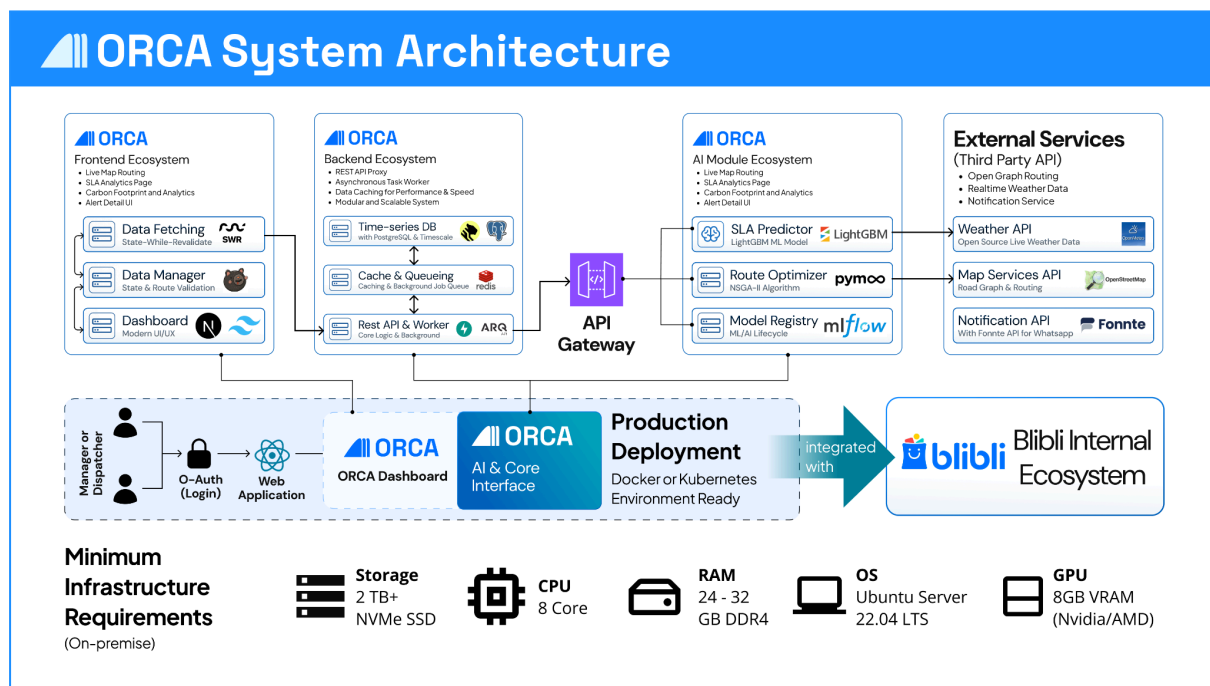


Fig. 1 Technical Architecture Diagram and System Workflow of ORCA

2.3. How does the solution work?

The input of the system consists of shipment data entering the ORCA platform via *REST API*, *Redis* queue, and dashboard polling. Each record carries operational data such as origin hub, destination, vehicle type, load, SLA deadline, and route distance (via *OSMnx* or *Haversine*).

In the processing stage, the engine executes four main steps. First, a *LightGBM* model predicts delay probability using 16 features, including weather severity from *Open-Meteo* and Indonesian calendar signals. Second, SLA risk is computed with urgency weighting and slack dampening based on remaining time. Third, *NSGA-II* performs route optimization by evaluating multiple alternatives across four objectives: time, cost, CO₂, and SLA risk, generating *Pareto-optimal* options. Fourth, the system calculates carbon emissions using the *GLEC v3.0* methodology.

The resulting insights are stored in the PostgreSQL database and delivered to the dashboard. Through this automated workflow, operators monitor a live dashboard where shipments are ranked by risk, allowing high-risk deliveries (above SLA Risk threshold) to trigger alerts via Fonnte and surface with SHAP-based explainable risk factors. Dispatchers receive route alternatives with clear trade-off comparisons, while sustainability teams gain access to aggregated carbon analytics, including daily trends, vehicle breakdowns, and total emissions, enabling efficient ESG reporting without manual intervention.

Impact & Outcome

3.1. Key Benefits of Adopting the Solution

Adopting ORCA delivers significant benefits for users and stakeholders across logistics operations. Operations managers gain early visibility into shipment risks before they become missed SLAs, allowing them to proactively address issues instead of reacting to customer complaints, which greatly reduces response time. Dispatchers benefit from data-driven decision-making, as ORCA provides route alternatives with measurable trade-offs across speed, cost, and carbon emissions, enabling more confident and optimized routing choices. Hub supervisors can identify local bottlenecks more clearly through *hub dwell time* and congestion insights, helping them resolve operational issues without being overwhelmed by broader network complexity. Sustainability officers gain automated carbon visibility, as each shipment is assigned a CO₂ estimate that is aggregated into ready-to-use analytics and ESG reports, eliminating manual tracking and reconciliation efforts. At the business level, ORCA improves SLA compliance through early risk detection, enhances fleet utilization through *multi-objective route optimization*, and strengthens sustainability accountability with transparent, shipment-level carbon reporting.

3.2. Shorts and Mid-Term Outcomes

In the short term (0-3 months), the operators gain instant visibility into shipment risks through automated dashboards, reducing manual monitoring efforts and enabling proactive intervention. Dispatchers can begin with making routing decisions based on data-driven insights. In the mid term (3-12 months), the Machine Learning model can improve in prediction accuracy, alert thresholds become better calibrated, and the route optimization becomes more reliable using real-world logistics patterns. Carbon analytics also become more mature, supporting more credible ESG reporting, while hub-level insights help identify recurring bottlenecks and structural inefficiencies across the network. In the longer term (12+ months), ORCA evolves into a core decision-support layer embedded in daily logistics operations. SLA performance improves across the network, fleet utilization becomes more efficient, and carbon reduction becomes a measurable operational target. The platform's real-time engine, predictive models, and optimization foundation also support future expansion into advanced capabilities such as automated decision support, real-time route re-optimization, predictive congestion detection, intelligent exception management, deeper integration with enterprise logistics systems, and adaptive machine learning models trained on continuously evolving operational data.

Innovation & Differentiation

4.1. What Makes Your Solution Different?

Unlike conventional logistics systems that optimize primarily for shortest distance or delivery speed, ORCA introduces a *carbon-aware optimization* approach that simultaneously balances SLA reliability, operational efficiency, and environmental impact in real time. The platform combines predictive AI, dynamic route optimization, shipment-level carbon analytics, and real-time *SLA risk scoring* into a unified decision-support ecosystem. ORCA's key differentiation lies in its ability to transform logistics operations from reactive management into proactive orchestration. The system predicts delivery disruptions before failures occur, enabling earlier intervention, smarter dispatch allocation, and congestion-aware decision-making. Route recommendations are evaluated not only based on distance and travel time, but also fuel efficiency, vehicle load, traffic conditions, and estimated carbon emissions. Unlike many global logistics platforms designed primarily for Western infrastructure, ORCA is optimized for Indonesia's highly fragmented and congestion-heavy logistics environment by integrating *Open-Meteo* weather data, Jabodetabek road network routing (*OSMnx/Stadia*), Indonesia calendar features (Lebaran, Ramadan, Harbolnas), and hub-level congestion metrics aligned with local last-mile conditions. This positions ORCA not merely as a routing tool, but as an AI-powered sustainability intelligence platform built for the future of resilient commerce [5][9].

4.2. Positioning of the Solution Compared to Existing Approaches or Products

Existing logistics platforms primarily focus on operational efficiency through static route optimization, fleet tracking, or delivery monitoring. While these systems improve visibility, most still operate reactively and rarely integrate predictive SLA intelligence with sustainability analytics into a unified platform. International logistics platforms such as FourKites and project44 provide supply chain visibility but are primarily designed for Western logistics infrastructure and lack localization for Indonesia's archipelagic geography, fragmented last-mile ecosystems, and highly dynamic traffic conditions. ORCA differentiates itself by integrating delay prediction, *carbon-aware optimization*, and real-time operational risk analysis into a single AI-driven decision engine. Rather than optimizing solely for speed or shortest distance, ORCA applies *multi-objective optimization* to balance delivery reliability, operational cost, fuel efficiency, and carbon impact simultaneously. In addition, ORCA provides shipment-level carbon analytics and predictive SLA scoring capabilities that support ESG accountability and sustainability reporting requirements for companies such as Blibli [9]. This positions ORCA as a next-generation logistics intelligence platform rather than a conventional routing solution.

Technical Approach

5.1. Main Solution

ORCA integrates five core technology layers: (1) a *LightGBM* binary classifier with sigmoid calibration for shipment delay prediction, trained using *Optuna* hyperparameter optimization (default 50 trials); (2) an *NSGA-II* multi-objective optimizer (*pymoo*) producing *Pareto-optimal* routes balancing ETA, fuel cost, CO₂ emissions, and SLA risk simultaneously; (3) a *GLEC v3.0* formula-based carbon engine computing per-shipment CO₂ using vehicle type, load weight, and route distance; (4) a *FastAPI* backend with *TimescaleDB* (*PostgreSQL 15* + *TimescaleDB* extension) for *Time-series* prediction storage and *Redis pub/sub* for real-time event streaming; and (5) a *Next.js* dashboard with *SWR*-based live polling for operational visibility. External integrations include *Open-Meteo* for real-time weather severity scoring and Fonnte for WhatsApp-based SLA breach alerts. The system supports both real-time *event-driven inference* via *Redis* subscriber and *REST API*-based on-demand prediction with *SHAP* explainability.

5.2. Technology Selection and Implementation

LightGBM was selected over *XGBoost* and *Random Forest* due to its superior handling of imbalanced binary classification (8.1% positive rate) and native support for categorical features. *CalibratedClassifierCV* with *5-fold sigmoid calibration* ensures well-calibrated probabilities ($ECE = 0.007$), critical for accurate *SLA risk scoring*. *Optuna TPE sampler* enables efficient hyperparameter search over a large search space without exhaustive grid search. *NSGA-II* via *pymoo* was chosen over Genetic Algorithms for its proven *Pareto-optimal* efficiency in logistics multi-objective problems. *TimescaleDB* extends *PostgreSQL* with time-series compression and hypertables, ideal for high-frequency shipment prediction writes. *Redis* provides sub-millisecond *pub/sub* for real-time event propagation and *LRU caching* of expensive predictions. *Open-Meteo* provides free, high-resolution weather forecasts without API key requirements, ensuring demo reliability. *SHAP* enables post-hoc explainability on each prediction for operator trust.

5.3. Solution Algorithm

ORCA employs two core algorithmic approaches to deliver its predictive and optimization capabilities. For delay prediction, a *LightGBM* classifier computes a *calibrated delay probability* using 16 features, including cyclical timing, *Open-Meteo* weather severity, hub delay rates, and Indonesia-specific calendar flags. *SHAP* values generate per-feature contribution scores surfaced in the UI to ensure explainability. A deterministic rule-based fallback activates automatically when the production model is unavailable, ensuring 100% risk score coverage across all active shipments. The SLA risk score is then calculated by applying slack dampening (*remaining/travel hours*) and urgency weighting to the delay probability. For route optimization, *NSGA-II* via *pymoo* simultaneously optimizes four competing objectives like ETA, fuel cost, CO₂ emissions, and SLA risk and returns a Pareto frontier of *non-dominated route alternatives for operator decision-making*. Carbon estimation

follows the *GLEC v3.0* formula based on distance, load weight, and vehicle-specific emission factors.

5.4. Primary Data or Input Used

ORCA leverages the publicly available *Olist Brazilian E-Commerce dataset* (comprising 96,470 orders from 2016 to 2018) as a training proxy. Raw files are stored locally in the data directory and excluded from Git due to their substantial size. To bridge the geographical gap, Indonesia specific calendar features such as Lebaran, Ramadan, and Harbolnas are mapped onto the timestamps to approximate local disruption patterns. However, variations in infrastructure and SLA practices limit its direct applicability, making the future integration of Bibli anonymized operational data essential for production deployment.

For real time inference, shipment events enter the platform through *REST API* or *Redis pub/sub*. These events are enriched with live weather severity scores from *Open-Meteo*, rolling hub performance metrics from *TimescaleDB*, Jabodetabek road network distances via *OSMnx*, and *GLEC v3.0* emission factors. To ensure the highest data quality and reliability, all features are versioned in a dedicated metadata configuration to prevent train and serve skew. Furthermore, the system utilizes expanding means with strict temporal ordering for all historical features to completely eliminate the risk of *data leakage* during the machine learning pipeline execution.

5.5. Security and Scalability Considerations

ORCA ensures data integrity through a two-tier *API security architecture*. Public endpoints are protected by *Bearer token authentication* with a strict rate limit of 120 requests per minute per IP, while internal ML inference routes are isolated via high-privilege tokens. All sensitive credentials are managed exclusively through environment variables, and production traffic is secured with origin-restricted CORS policies.

To support future growth, the stateless *FastAPI* backend is designed for horizontal scaling behind a load balancer. The system handles high-frequency operational data using *TimescaleDB hypertables* for optimized time-series storage. Performance is further enhanced by *Redis* caching, which stores route optimization results and hub metrics to reduce database load. Additionally, the compute-intensive *NSGA-II* optimization is optimized through *route fingerprinting* and caching, ensuring that the platform remains responsive as shipment volumes increase and the distribution network expands.

Implementation Feasibility

6.1. Invention Status

The invention is currently at the Prototype stage. To validate our proposal ahead of the submission deadline, our team proactively built a functional baseline system featuring a *LightGBM* delay prediction model, an *NSGA-II* route optimizer, a *GLEC v3.0* carbon calculator, and a live *Next.js* dashboard. All components are fully containerized via *Docker*, demonstrating the technical readiness and feasibility of our architecture before advancing to the subsequent phases of the competition.

6.2. Is the Innovation Realistic to Build?

Yes, ORCA is highly realistic to implement. Our team has already proven its technical feasibility by developing a working prototype during this proposal phase using entirely open-source, production-ready technologies like *LightGBM*, *FastAPI*, *TimescaleDB*, and *Next.js*. This ensures no restrictive proprietary dependencies and seamless deployment within Blibli's infrastructure. The architecture relies on standard *REST APIs* and event-driven patterns, making it highly compatible with existing Transport Management Systems. Furthermore, the AI components are lightweight and execute efficiently on commodity hardware. Because we have successfully validated the core routing algorithms and predictive pipelines using proxy data, the primary remaining step is ingesting real operational data. Our team possesses the exact blend of machine learning and backend engineering expertise required to transition this validated prototype into a fully operational solution by the final event.

6.3. Development Stages

Stage 1: Baseline Prototype (Current Phase, Complete)

We delivered a functional ORCA prototype utilizing proxy data. It features a *LightGBM* delay prediction model with *SHAP* explainability, an *NSGA-II* route optimizer, and *GLEC v3.0* carbon accounting. The *FastAPI*, *Redis pub/sub*, and *TimescaleDB* architecture powers a *Next.js* dashboard covering SLA alerts, optimization, carbon analytics, and hub monitoring.

Stage 2: Semifinal Refinement (June)

We will refine the dashboard to include SLA risk heatmaps, fleet metrics, and executive views. Utilizing tech tutorials, we will calibrate thresholds and produce a video demonstration showcasing these analytical layers across seeded Jabodetabek logistics scenarios.

Stage 3: Enterprise Integration (July to August)

During the Company Visit and Mentoring phases, we will ingest anonymized Blibli operational data. We will retrain machine learning pipelines on localized patterns and connect the architecture to live GPS, TMS, ERP, and traffic API feeds.

Stage 4: Pilot Deployment (September Onwards)

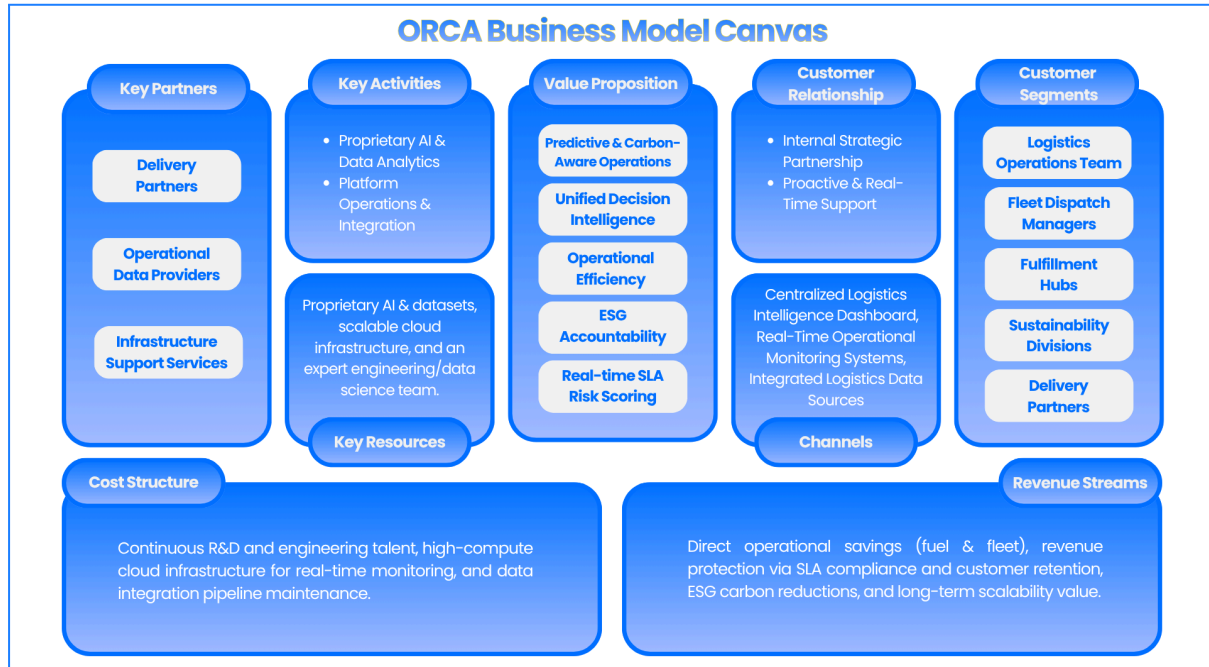
Following the Final Event, the MVP will be deployed at a Blibli fulfillment hub. We will monitor SLA compliance, detect operational bottlenecks, and track carbon outcomes to establish a Jabodetabek rollout plan.

6.4. Business Model and Sustainability

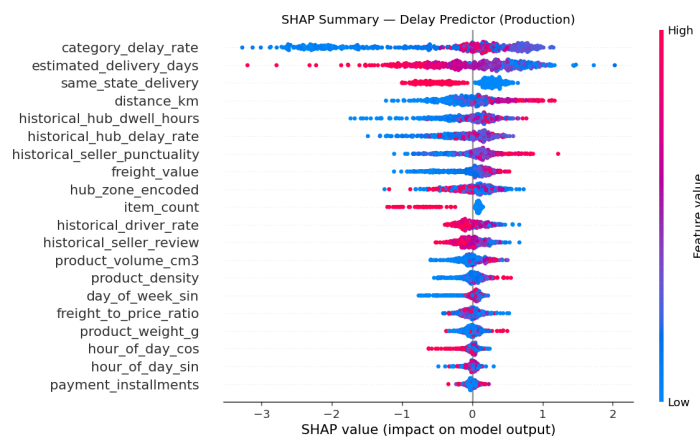
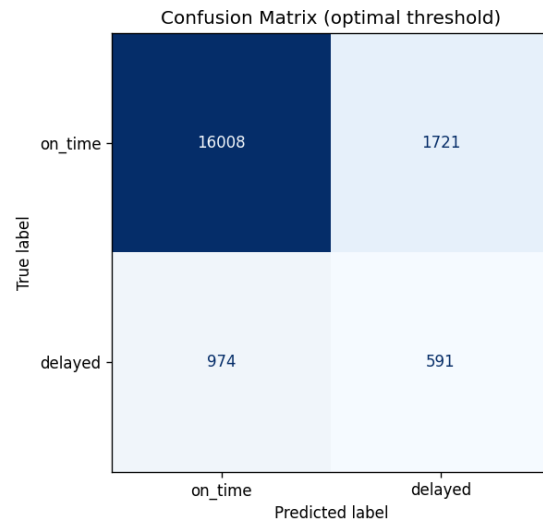
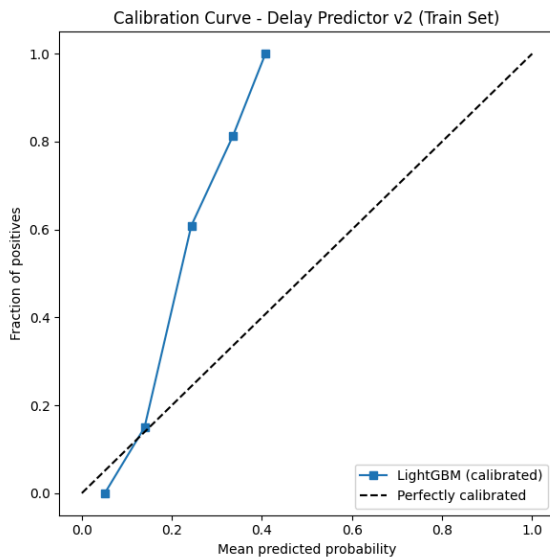
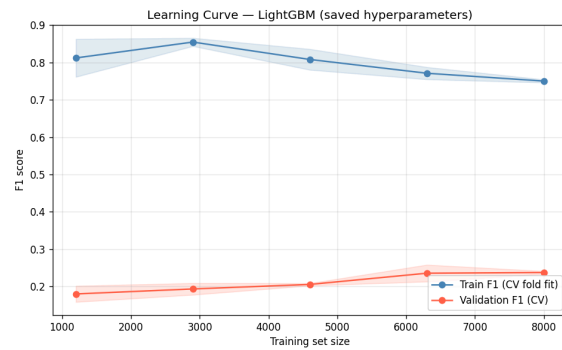
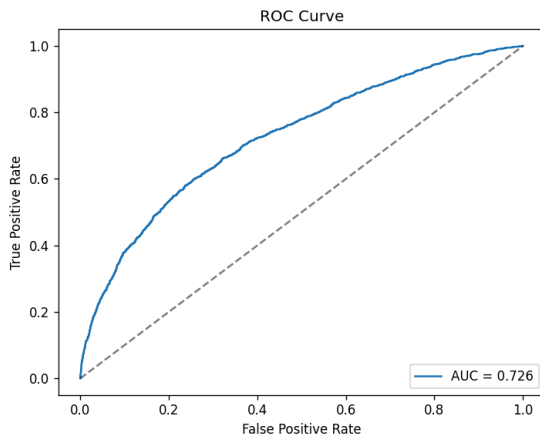
ORCA is an enterprise AI logistics platform designed to help Blibli transform its nationwide logistics operations from reactive execution into predictive, carbon-aware decision-making. Serving customer segments like logistics teams, dispatch managers, fulfillment hubs, and sustainability divisions, the platform fosters proactive, real-time support (customer relationships) via an integrated logistics intelligence dashboard (channels). Its core value proposition lies in integrating AI-powered delay prediction, route optimization, real-time SLA risk scoring, and shipment-level carbon analytics into a unified decision intelligence platform to maximize operational efficiency and ESG accountability. To drive this, key activities focus on proprietary AI analytics and platform integration, powered by key resources like scalable cloud infrastructure, datasets, and an expert engineering/data science team. ORCA collaborates with delivery partners, operational data providers, and infrastructure support services as key partners. As an internal solution, its cost structure is driven by continuous R&D, engineering talent, and high-compute cloud infrastructure maintenance. Instead of direct monetization, revenue streams are realized through value capture: direct operational savings in fuel and fleet, revenue protection via SLA compliance, customer retention, and measurable carbon reduction, ensuring long-term scalability.

Attachment & Reference

Appendix 1. Business Model Canvas of ORCA



Appendix 2. Accuracy Model

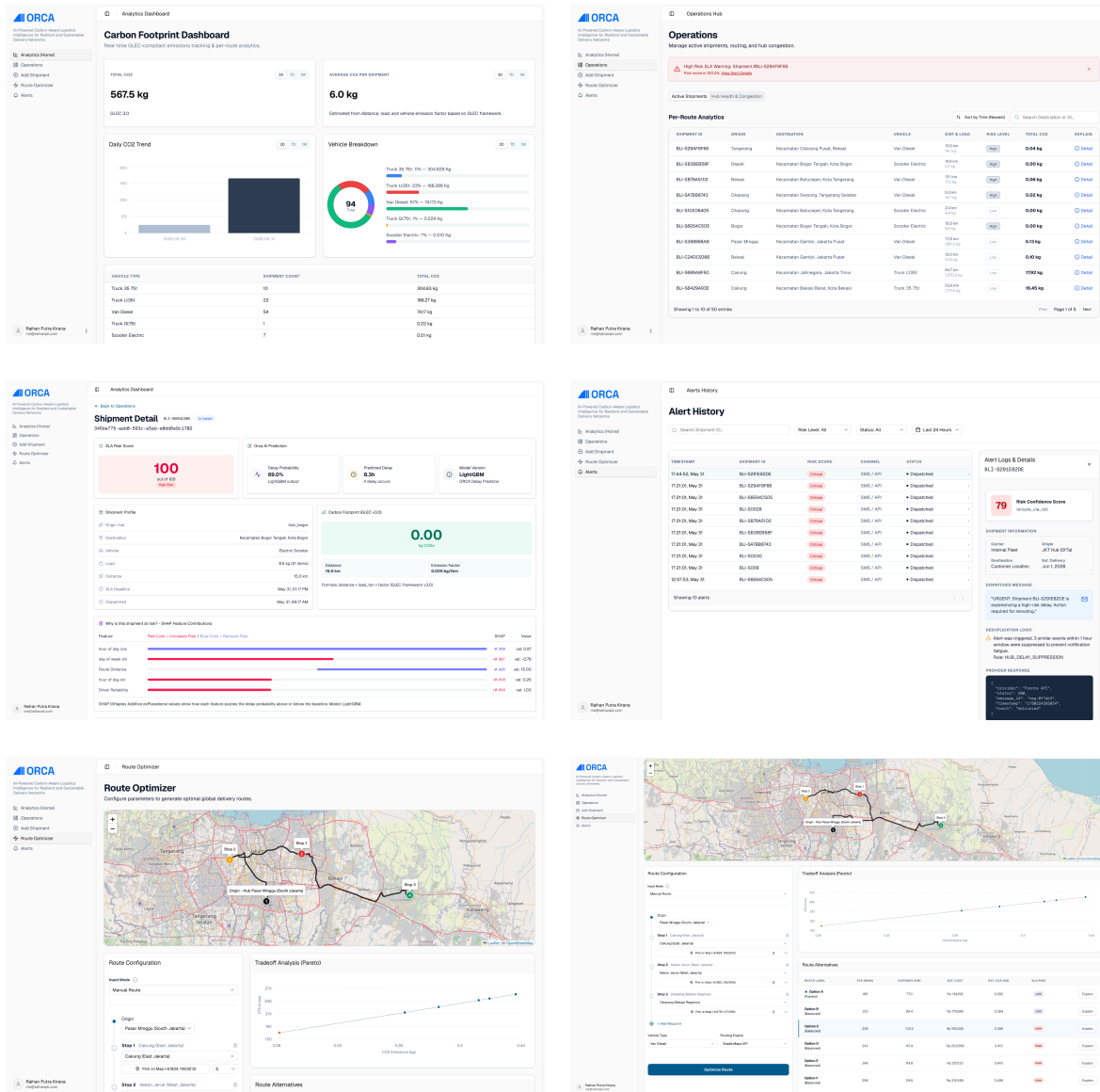


Appendix 3. Web Application Prototype

Github Repository: <https://github.com/raihanpka/orca-ai-open-innovation>

Demo Video Link: [Demo Prototype Video](#)

Web can be accessed via: <https://orca-ai.thelunareix.my.id>



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